Exercises from Ch 2: Rings, Lang

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Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Exercise 0.1. (Exercise 1) Suppose that $1 \neq 0$ in A. Let S be a multiplicative subset of A not containing 0. Let $\mathfrak{p}$ be a maximal element in the set of ideals of A whose intersection with S is empty. Show that $\mathfrak{p}$ is prime.

Proof. Denote the set of ideals of A whose intersection with S is empty by

$$J(A) = \{I \subseteq A \text{ ideal} : \ I \cap S = \varnothing\}.$$

Let $\mathfrak{p} \in J(A)$ be maximal. Hence, for all $I \in J(A) \setminus \mathfrak{p}$, $I \subseteq \mathfrak{p}$. By definition, $\mathfrak{p} \cap S = \varnothing$. Let $a, b \in A$ such that $ab \in \mathfrak{p}$. If $a \in \mathfrak{p}$, we are done. Thus, assume $a \notin \mathfrak{p}$.
<u>Case 1 ($a \in S$)</u>: By way of contradiction, suppose $b \notin \mathfrak{p}$. We form the ideal

$$\mathfrak{p} + (b) = \{x + rb : \ x \in \mathfrak{p}, \, r \in A\}.$$

Then, $\mathfrak{p} \subseteq \mathfrak{p} + (b)$. But $\mathfrak{p} + (b) \notin J(A)$ by the maximality of $\mathfrak{p} \in J(A)$. Thus,

$$(\mathfrak{p} + (b)) \cap S \neq \varnothing.$$

Then, there exists some $s \in S$ such that $s = x + rb$. Choose $r = a \in A$. Thus, $s = x + ab \in \mathfrak{p}$ since $x \in \mathfrak{p}$ and $ab \in \mathfrak{p}$ by assumption. Thus, $s \in \mathfrak{p}$. But $s \in S$ and $\mathfrak{p} \cap S = \varnothing$. Contradiction! Therefore, $b \in \mathfrak{p}$.
<u>Case 2 ($a \notin S$)</u>: Hence, $a \notin \mathfrak{p}$ and $a \notin S$, thus $a \in A$ strictly. Consider the ideal

$$\mathfrak{p} + (a) = \{x + ra : \ x \in \mathfrak{p}, \, r \in A\}.$$

Then, since $a \notin \mathfrak{p}$, $\mathfrak{p} \subseteq \mathfrak{p} + (a)$. But $\mathfrak{p} + (a) \notin J(A)$ by the maximality of $\mathfrak{p} \in J(A)$. Thus,

$$(\mathfrak{p} + (a)) \cap S \neq \varnothing.$$

Then, there exists some $s \in S$ such that $s = x + ra$ with $r = b \notin \mathfrak{p}$. Thus, $x + ba \in \mathfrak{p} \Rightarrow s \in \mathfrak{p}$. but $s \in S$ and $\mathfrak{p} \cap S = \varnothing$. Contradiction! Thus, $b \in \mathfrak{p}$.

□

Exercise 0.2. (Exercise 2) Let $f \colon A \to A'$ be a surjective homomorphism of rings, and assume that A is local, $A \neq 0$. Show that A' is local.

Proof. Since f is surjective, by the First Isomorphism Theorem, $A' \cong A/\ker f$. Let $\mathfrak{m} \subseteq A$ be the unique maximal ideal (since A is local by assumption). Since $A' \neq 0$, we have $\ker f \subsetneq A$ (proper ideal) which implies $\ker f \subseteq \mathfrak{m}$ since $\mathfrak{m}$ is maximal. Maximal ideals of the quotient ring $A/\ker f$ are in bijection with maximal ideals $\mathfrak{n} \subseteq A$ such that $\ker f \subseteq \mathfrak{n}$ via $\mathfrak{n} \mapsto \mathfrak{n}/\ker f$. Since A has exactly one maximal ideal $\mathfrak{m}$ and $\ker f \subseteq \mathfrak{m}$, it follows that $A/\ker f$ has exactly one maximal ideal $\mathfrak{m}/\ker f$. Define

$$\varphi \colon A/\ker f \to A'$$

by $\varphi(x + \ker f) = f(x)$ for all $x \in A$. Hence, $\varphi(\mathfrak{m}/\ker f) = \{f(x) \colon x \in \mathfrak{m}\} = f(\mathfrak{m})$. Therefore, under $A' \cong A/\ker f$, $\mathfrak{m}/\ker f$ corresponds to $f(\mathfrak{m}) \subseteq A'$, i.e. $\mathfrak{m}/\ker f$ and $f(\mathfrak{m})$ are the "same" ideal just in different "languages". Thus, A' has a unique maximal ideal (and commutative by assumption), A' is local.

□

Exercise 0.3. (Exercise 3)
Let $\mathfrak{p}$ be a prime ideal of A. Show that $A_\mathfrak{p}$ has a unique maximal ideal, consisting of all elements a/s with $a \in \mathfrak{p}$ and $s \notin \mathfrak{p}$.

Proof. We recall that, by definition, $A_\mathfrak{p} = \{a/s \colon a \in A,\ s \notin \mathfrak{p}\}$ is the local ring of A at $\mathfrak{p}$. We first prove the following corollary.

Corollary 0.1. Let $x/s \in A_\mathfrak{p}$. Then x/s is not a unit if and only if $x \in \mathfrak{p}$. (Equivalently, x/s is a unit if and only if $x \notin \mathfrak{p}$.)

Proof. ($\Longrightarrow$) We proceed via the contrapositive. Suppose $x \notin \mathfrak{p}$. Consider $x/1 \in A_\mathfrak{p}$. Since $\mathfrak{p}$ is prime, $x \notin \mathfrak{p}$ implies that $x \in S = A \setminus \mathfrak{p}$. Every element of S is invertible in the localization, since if $x \in S$, then $1/x \in A_\mathfrak{p}$ and $x/1 \in A_\mathfrak{p}$ since $S \subseteq A$, thus $(x/1)(1/x) = 1/1$. Then if $x \notin \mathfrak{p}$, $x/s = (x/1)(1/s) \in A_\mathfrak{p}$ is a unit since a product of units is a unit.
($\Longleftarrow$) Let $x \in \mathfrak{p}$. By way of contradiction, suppose $x/s \in A_\mathfrak{p}$ is a unit. Then there exists some $y/t \in A_\mathfrak{p}$ such that $(x/s)(y/t) = xy/st = 1/1$. Then, by equivalence in the localization, there exists some $s' \in S$ such that $s'(xy - st) = 0$. Then $s'xy = s'st$. Since $\mathfrak{p} \subseteq A$ is an ideal and $x \in \mathfrak{p}$, $s'xy \in \mathfrak{p}$. But $s'st \notin \mathfrak{p}$. Contradiction!
Therefore, $x/s \in A_\mathfrak{p}$ is a unit if and only if $x \notin \mathfrak{p}$.

□

We return to the proof of the exercise. By the corollary, the set of nonunits of $A_\mathfrak{p}$ is $\mathfrak{m} = \{a/s \colon a \in \mathfrak{p},\ s \notin \mathfrak{p}\} \subseteq A_\mathfrak{p}$. Observe that this is certainly an ideal since multiplication by a unit does not make a nonunit a unit. We show that $\mathfrak{m}$ is maximal. Suppose that there exists another ideal $I \subseteq A_\mathfrak{p}$ such that $\mathfrak{m} \subsetneq I$. Then there exists some $x/s \in I$ such that $x/s \notin \mathfrak{m}$. Thus, it follows that x/s is a unit. Hence, $I = A_\mathfrak{p}$. Therefore, $\mathfrak{m} \subseteq A_\mathfrak{p}$ is maximal. It is unique since every maximal ideal consists of only nonunits and $\mathfrak{m}$ is precisely all of the nonunits. Thus, every maximal ideal is equivalent to $\mathfrak{m}$, and therefore is unique.
Therefore, $A_\mathfrak{p}$ has a unique maximal ideal consisting of a/s such that $a \in \mathfrak{p}$ and $s \notin \mathfrak{p}$.

□

Exercise 0.4. (Exercise 4) Let A be a principal ring and S a multiplicative subset with $0 \notin S$. Show that $S^{-1}A$ is principal.

Proof. Define $f : A \to S^{-1}A$ by the canonical localization map $f(a) = a/1$. Let $J \subseteq S^{-1}A$ be an ideal and define

$$I = f^{-1}(J) = \{x \in A : \ x/1 \in J\} \subseteq A.$$

<u>Case 1:</u> ($I \cap S \neq \varnothing$) Let $s \in I \cap S \Rightarrow s/1 \in J$. By definition of localization, $s/1$ is a unit in $S^{-1}A$ with $(s/1)^{-1} = 1/s$. Hence, since J is an ideal that contains a unit $\underbrace{(s/1)}_{\in J}\underbrace{(1/s)}_{\in S^{-1}A} = 1/1 = 1 \in J$. But if $1 \in J$, then $J = S^{-1}A$. Hence, J is principal with $J = S^{-1}A = (1/1) = (1)$.
<u>Case 2:</u> ($I \cap S = \varnothing$) WTS $J = S^{-1}I$. ($\subseteq$) If $x/s \in J$, then $x/1 = (x/s)(s/1) \in J \Rightarrow x \in I$. Hence, $x/s \in S^{-1}I$ and thus $J \subseteq S^{-1}I$. ($\supseteq$) If $x \in I$, then $x/1 \in J$. Since J is an ideal, for all $s \in S$, we have

$$\underbrace{(x/1)}_{\in J}\underbrace{(1/s)}_{\in S^{-1}A} = x/s \in J.$$

Thus, $J \supseteq S^{-1}I$. Finally, since A is principal, $I = (a)$. Therefore, $J = S^{-1}I = S^{-1}(a) = (a/1)$. Hence, J is principal generated by $a/1$. Therefore, every ideal $J \subseteq S^{-1}A$ is principal and thus $S^{-1}A$ is principal. □

Exercise 0.5. (Exercise 5)
Let A be a factorial ring and S a multiplicative subset with $0 \notin S$. Show that $S^{-1}A$ is factorial, and that the prime elements of $S^{-1}A$ are of the form up with primes p in A such that $(p) \cap S$ is empty, and units u in $S^{-1}A$.

Proof. **NEED TO WORK OUT THE PROOF FULLY!!! SEE NOTES!!!**
Suppose $x/s \in S^{-1}A$ is a unit. Then $y/t \in S^{-1}A$ such that $(x/s)(y/t) = xy/st = 1/1$. Thus, there exists some $s_1 \in S$ such that $s_1(xy - st) = 0$. Since A is factorial and $S \subseteq A$, A is entire and therefore $s_1 = 0$ or $xy - st = 0$, but $0 \notin S$. Therefore $xy - st = 0 \Rightarrow xy = st$. Hence, $xy \in S$.
$p/1 \in S^{-1}A$ is prime in the localization if it is not a unit and if whenever ne can write $p/1 = (x/s)(y/t)$ then one of $x/s, y/t$ is a unit. Hence we showed it is not a unit if $(p) \cap S = \varnothing$. Assume that

$$\begin{aligned} p/1 &= (x/s)(y/t) \\ &= xy/st \end{aligned}$$

thus there exists some $s' \in S$ such that $s'(pst - xy) = 0 \Rightarrow pst = xy$. Hence $xy \in (p)$. Since $(p) \subseteq A$ is prime, $x \in (p)$ or $y \in (p)$.

□

Exercise 0.6. (Exercise 6) Let A be a factorial ring and p a prime element. Show that the local ring $A_{(p)}$ is principal.

Proof. If A is also principal, then we are done by Exercise 4. Thus, suppose A is only factorial. Let p be an irreducible (prime) element of A. Thus, $(p) \subseteq A$ is a prime ideal. Set $S = A \setminus (p)$. Hence,

$$S^{-1}A = A_{(p)} = \{x/s : \ x \in A, \ s \notin (p)\}.$$

Let $\mathfrak{m} := \{x/s \in A_{(p)} : \ x \in (p)\}$. By definition, if $x \notin (p)$, then $x \in S$, so $x/1$ is invertible in the localization (since $1 \in A$, $x \in S$, $1/x \in A_{(p)}$). Thus, x/s is invertible. Now, if $x \in (p)$, then no matter what $y/t \in A_{(p)}$ we multiply by, we obtain $xy/st \in (p)A_{(p)} = \{pa/s : \ a \in A, \ s \notin (p)\} = (p/1) \subseteq A_{(p)}$, which is never equal to 1. Thus, it is not invertible, so $A \setminus \mathfrak{m} = A_{(p)}^{\times}$ (units of $A_{(p)}$). Thus, $A_{(p)}/\mathfrak{m}$ is a field. Therefore, $\mathfrak{m}$ is the unique maximal ideal of $A_{(p)}$, hence $A_{(p)}$ is local.
Finally, if $x \in (p)$, we write $x = pk$. Then

$$x/s = (p/1) \cdot (k/s),$$

so every element of $\mathfrak{m}$ is a multiple of $p/1$. Thus, $\mathfrak{m} = (p/1)$.
Now we must show all ideals are principal. Let I be a nonzero ideal of $A_{(p)}$. Pick an element $x/s \in I$ with $x \neq 0$ having the smallest exponent of p in its factorization $x = up^n$ (where u is a unit in A). Then

$$x/s = (p/1)^n \cdot (u/s),$$

and u/s is a unit in $A_{(p)}$, since $u \notin (p)$. Hence $(x/s) = (p/1)^n$. If $y/t \in I$, then $y = u'p^m$ for some $m \geq n$ by minimality of n, and thus

$$y/t = (p/1)^m \cdot (u'/t) \in (p/1)^n.$$

Therefore, $I = (p/1)^n$, proving that every ideal of $A_{(p)}$ is principal.
Thus, $A_{(p)}$ is principal. □

Exercise 0.7. (Exercise 7) Let A be a principal ring $a_1, \ldots, a_n$ non-zero elements of A. Let $(a_1, \ldots, a_n) = (d)$. Show that d is a greatest common divisor for the a_i $(i = 1, \ldots, n)$.

Proof. By construction, $a_i \in (a_1, \ldots, a_n) = (d)$
$\left(\text{i.e. } a_i = \sum_{i=1}^{n} x_i a_i, \text{ w/ } x_j = 0 \text{ if } j \neq i \text{ and } x_j = 1 \text{ if } j = i\right)$. Thus, there exists $b_i \in A$ such that $a_i = b_i d$ for all $1 \leq i \leq n$. Hence, $d \big| a_i$ for all $1 \leq i \leq n$. Now, suppose that there exists $c \in A$ such that $c \big| a_i$ for all $1 \leq i \leq n$. Then, for all i, there exists $y_i \in A$ such that $a_i = y_i c$. Therefore, $a_i \in (c)$ (since if $x_1 a_1 + \cdots + x_n a_n \in (a_1, \ldots, a_n)$, then by the above $x_1 y_1 c + \cdots + x_n y_n c \in (c)$, clearly). But by assumption, $(a_1, \ldots, a_n) = (d)$, whence $(d) \subseteq (c)$. Thus, there exists $z \in A$ such that $d = zc \Rightarrow c \big| d$. Therefore any common divisors of the a_i's divides d. Hence, d is the greatest common divisor of a_i, $i = 1, \ldots, n$.

□

Exercise 0.8. (Exercise 8)
Let p be a prime number, and let A be the ring $\mathbb{Z}/p^r\mathbb{Z}$ ($r = \text{integer} \geq 1$). Let G be the group of units in A, i.e. the group of integers prime to p, modulo p^r. Show that G is cyclic, except in the case when

$$p = 2, \quad r \geq 3$$

in which case it is of type $(2, 2^{r-2})$. [*Hint*: In the general case, show that G is the product of a cyclic group generated by $1+p$, and a cyclic group of order $p-1$. In the exceptional case, show that G is the product of the group $\{\pm 1\}$ with the cyclic group generated by the residue class of $5 \mod 2^r$.]

Proof. Let p be an odd prime and $r \geq 1$. Consider $G = A^\times = (\mathbb{Z}/p^r\mathbb{Z})^\times$ with $|G| = \varphi(p^r) = p^r - p^{r-1} = p^{r-1}(p-1)$. Define

$$\pi \colon (\mathbb{Z}/p^r\mathbb{Z})^\times \to (\mathbb{Z}/p\mathbb{Z})^\times$$

by $x \mapsto x \mod p$. Thus, $\ker \pi = \{1 + pk \colon k \in \mathbb{Z}/p^{r-1}\mathbb{Z}\}$. Hence, $|\ker \pi| = p^{r-1}$. We show that $\ker \pi = \langle 1+p \rangle$. Thus,

$$(1+p)^{p^{r-1}} = \sum_{k=0}^{p^{r-1}} \binom{p^{r-1}}{k} p^k = 1 + p^{r-1} \cdot p + \binom{p^{r-1}}{2} p^2 + \cdots \mod p^r$$

Hence, the second term, $p \cdot p^{r-1} = p^r \equiv 0 \mod p^r$. Similarly, for all $k \geq 2$, we have $p^r \big| \binom{p^{r-1}}{k} p^k$. Thus, we see $(1+p)^{p^{r-1}} \equiv 1 \mod p^r$. Now suppose there exists some $m < p^{r-1}$ such that $(1+p)^m \equiv 1 \mod p^r$. Then, the second term in the binomial expansion is pm and thus we have $p^r \big| pm \Rightarrow p^{r-1} \big| m$ and thus since $m < p^{r-1}$, this is absurd and thus p^{r-1} is the order of $1+p$.
Thus, $|\ker \pi| = |\langle 1+p \rangle| = p^{r-1}$. Therefore, $\ker \pi = \langle 1+p \rangle$. By the First Isomorphism Theorem,

$$G/\ker \pi \cong \pi(G) \Rightarrow (\mathbb{Z}/p^r\mathbb{Z})^\times / \langle 1+p \rangle \cong (\mathbb{Z}/p\mathbb{Z})^\times$$

Hence, $G/\ker \pi$ is cyclic, since $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic from the fact that it is the multiplicative group of a field, and $|G/\ker \pi| = \varphi(p) = p-1$.
In the general case, let $x \ker \pi \in G/\ker \pi$. Then $x^{p-1} \in \ker \pi$. Thus, since $|\ker \pi| = p^{r-1}$, $|x^{p-1}| \Big| |\ker \pi| = p^{r-1}$. Since $(|\ker \pi|, |G/\ker \pi|) = (p^{r-1}, p-1) = 1$, we have $|x^{p-1}| = p^{r-1}$ and therefore $|x| = p^{r-1}(p-1) = p^r - p^{r-1} = |G|$. Hence, G is cyclic generated by x.
We now consider the case when $p = 2$ and $r \geq 3$. Then we immediately see that our surjective homomorphism π, in this case, results in a trivial quotient with G ($\ker \pi = G$ since $(\mathbb{Z}/2\mathbb{Z})^\times = \{1\}$) so we have to proceed differently. First, we note that $|G| = |(\mathbb{Z}/2^r\mathbb{Z})^\times| = \varphi(2^r) = 2^{r-1}$. Consider the cyclic subgroup $K = \{\pm 1\} = \{1, 2^r - 1\} \leqslant G$. This is a cyclic group of order 2. We now wish to find some other cyclic subgroup $H \leqslant G$ of order 2^{r-2}. We claim that $5 \in G$ has order 2^{r-2}. That is

$$5^{2^{r-2}} \equiv 1 \mod 2^r$$

Consider the binomial expansion,

$$5^{2^{r-2}} = (1+2^2)^{2^{r-2}} = \sum_{k=0}^{2^{r-2}} \binom{2^{r-2}}{k} \left(2^2\right)^k = 1 + 2^{r-2} \cdot 2^2 + (\text{higher terms})$$

All higher terms contain factors $2^{2k}\binom{2^{r-2}}{k} \geq 2^r$ and therefore vanish modulo 2^r. Likewise, the second term, $2^{r-2} \cdot 2^2 = 2^r \equiv 0 \mod 2^r$. Thus, $5^{2^{r-2}} \equiv 1 \mod 2^r$. By a similar argument contained in the general case, one can check that if $m < 2^{r-2}$ and $5^m \equiv 1 \mod 2^r$, then $2^{r-2} \big| m$ which gives a contradiction to the inequality $m < 2^{r-2}$. Thus, let $H = \langle 5 \mod 2^r \rangle \leqslant G$. Hence, $|H| = 2^{r-2}$. Observe that $K \cap H = \{1\}$ since $5^k \equiv -1 \mod 2^r$ has no solutions, $0 \leq k \leq 2^{r-2}$. We see this observation by supposing it had a solution, thus $5^{2k} \equiv 1 \mod 2^r$ which implies that $2^{r-2} \big| 2k \Rightarrow 2^{r-3} \big| k$. Thus, we would require that $5^{2^{r-3}} \equiv 1 \mod 2^r$. But using the binomial expansion, we see that $5^{2^{r-3}} \equiv 1 + 2^{r-1} \not\equiv 1 \mod 2^r$ (or by the argument showing the order is 2^{r-2}). Furthermore, $|K| \cdot |H| = 2 \cdot 2^{r-2} = 2^{r-1} = |G|$. It then follows that

$$G \cong K \times H \Rightarrow (\mathbb{Z}/2^r\mathbb{Z})^\times \cong \{\pm 1\} \times \langle 5 \mod 2^r \rangle$$

Hence, since G is abelian (clear), it is of type $(2, 2^{r-2})$ and is not cyclic since no element has order 2^{r-1} (maximum order is $|5| = 2^{r-2}$).

$\square$

Dedekind rings

Prove the following statements about a Dedekind ring $\mathfrak{o}$. To simplify terminology, by an **ideal** we shall mean non-zero ideal unless otherwise specified. We let K denote the quotient field of $\mathfrak{o}$.

Exercise 0.9. (Exercise 17) As for the integers, we say that $\mathfrak{a} \mid \mathfrak{b}$ ($\mathfrak{a}$ **divides** $\mathfrak{b}$) if there exists an ideal $\mathfrak{c}$ such that $\mathfrak{b} = \mathfrak{a}\mathfrak{c}$. Prove:

a) $\mathfrak{a} \mid \mathfrak{b}$ if and only if $\mathfrak{b} \subseteq \mathfrak{a}$.

b) Let $\mathfrak{a}, \mathfrak{b}$ be ideals. Then $\mathfrak{a} + \mathfrak{b}$ is their greatest common divisor. In particular, $\mathfrak{a}, \mathfrak{b}$ are relatively prime if and only if $\mathfrak{a} + \mathfrak{b} = \mathfrak{o}$.

Proof. a) Note that $\mathfrak{a}, \mathfrak{b}$ are non-zero ideals of the Dedekind ring $\mathfrak{o}$.

($\Rightarrow$) Assume $\mathfrak{a} \mid \mathfrak{b}$. By definition this means there exists an ideal $\mathfrak{c}$ such that

$$\mathfrak{b} = \mathfrak{a}\mathfrak{c}.$$

Since $\mathfrak{c} \subseteq \mathfrak{o}$, we have $\mathfrak{a}\mathfrak{c} \subseteq \mathfrak{a}$. Thus $\mathfrak{b} \subseteq \mathfrak{a}$.

($\Leftarrow$) Conversely, assume $\mathfrak{b} \subseteq \mathfrak{a}$. Since $\mathfrak{o}$ is Dedekind, every non-zero ideal has a unique factorization into prime ideals. Namely, write

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\alpha_{\mathfrak{p}}} \quad \text{and} \quad \mathfrak{b} = \prod_{\mathfrak{p}} \mathfrak{p}^{\beta_{\mathfrak{p}}}$$

Observe that for all $\mathfrak{p}$, we have $\mathfrak{p}^m \subseteq \mathfrak{p}^n$ if and only if $m \geq n$. Then $\mathfrak{b} \subseteq \mathfrak{a} \Rightarrow \beta_{\mathfrak{p}} \geq \alpha_{\mathfrak{p}}$ for all $\mathfrak{p}$. Define another ideal,

$$\mathfrak{c} = \prod_{\mathfrak{p}} \mathfrak{p}^{\beta_{\mathfrak{p}} - \alpha_{\mathfrak{p}}}$$

Note that $\beta_{\mathfrak{p}} - \alpha_{\mathfrak{p}} \geq 0$ for all $\mathfrak{p}$ and thus $\mathfrak{c} \subseteq \mathfrak{o}$ (since the product of ideals is an ideal). Then,

$$\mathfrak{a}\mathfrak{c} = \left(\prod_{\mathfrak{p}} \mathfrak{p}^{\alpha_{\mathfrak{p}}}\right)\left(\prod_{\mathfrak{p}} \mathfrak{p}^{\beta_{\mathfrak{p}} - \alpha_{\mathfrak{p}}}\right) = \prod_{\mathfrak{p}} \mathfrak{p}^{\beta_{\mathfrak{p}}} = \mathfrak{b}$$

Hence, $\mathfrak{b} = \mathfrak{a}\mathfrak{c} \Rightarrow \mathfrak{a} \big| \mathfrak{b}$.

b) The gcd of ideals is analogous to the definition of gcd in the integers:

> An ideal $\mathfrak{d}$ is the $\gcd(\mathfrak{a}, \mathfrak{b})$ if $\mathfrak{d} \big| \mathfrak{a}$ and $\mathfrak{d} \big| \mathfrak{b}$; if $\mathfrak{c}$ is any ideal such that $\mathfrak{c} \big| \mathfrak{a}$ and $\mathfrak{c} \big| \mathfrak{b}$, then $\mathfrak{c} \big| \mathfrak{d}$.

We claim that $\gcd(\mathfrak{a}, \mathfrak{b}) = \mathfrak{a} + \mathfrak{b}$.

We have $\mathfrak{a} \subseteq \mathfrak{a} + \mathfrak{b}$ and $\mathfrak{b} \subseteq \mathfrak{a} + \mathfrak{b}$. Therefore, by part (a), $\mathfrak{a} + \mathfrak{b} \big| \mathfrak{a}$ and $\mathfrak{a} + \mathfrak{b} \big| \mathfrak{b}$. Now suppose there exists an ideal $\mathfrak{c}$ such that $\mathfrak{c} \big| \mathfrak{a}$ and $\mathfrak{c} \big| \mathfrak{b}$. Then, again by part (a), $\mathfrak{a} \subseteq \mathfrak{c}$ and $\mathfrak{b} \subseteq \mathfrak{c}$. We want to show $\mathfrak{a} + \mathfrak{b} \subseteq \mathfrak{c}$. But if $z = x + y \in \mathfrak{a} + \mathfrak{b}$, then $x \in \mathfrak{a} \subseteq \mathfrak{c}$ and $y \in \mathfrak{b} \subseteq \mathfrak{c}$. Since ideals are closed under addition, $z = x + y \in \mathfrak{c}$. Hence,

$$\mathfrak{a} + \mathfrak{b} \subseteq \mathfrak{c}$$

Then, by utilizing part (a), again, we have $\mathfrak{c} \big| \mathfrak{a} + \mathfrak{b}$. Therefore, $\mathfrak{a} + \mathfrak{b} = \gcd(\mathfrak{a}, \mathfrak{b})$.

The particular case mentioned follows immediately since if $\gcd(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ (unit ideal, the whole Dedekind ring), then by what we have shown $\gcd(\mathfrak{a}, \mathfrak{b}) = \mathfrak{a} + \mathfrak{b} = \mathfrak{o}$.

The converse is trivial.

$\square$